

Air University



OOP (Lab)

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Assignment No. 03**Question 01****Code****Row (1 – 23)**

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  class Publication
6  {
7  private:
8      string title;
9      float price;
10
11 public:
12     void setdata(string t, float p)
13     {
14         title = t;
15         price = p;
16     }
17     void getdata()
18     {
19         cout << "\nTitle: " << title << endl;
20         cout << "Price: " << price << "$" << endl;
21     }
22 };
23
```

Row (24 - 51)

```
24  class Book : public Publication
25  {
26  private:
27      int page_count;
28
29  public:
30      void setdata(int p_c, string t, float p)
31      {
32          Publication ::setdata(t, p);
33          page_count = p_c;
34      }
35      void getdata()
36      {
37          string title;
38          float price;
39          cout << "\n--- Enter Book Details ---" << endl;
40
41          cout << "Enter title of the book: ";
42          getline(cin, title);
43
44          cout << "Enter Price of the book: ";
45          cin >> price;
46          while (price < 0)
47          {
48              cout << "Kindly Enter Positive Price: ";
49              cin >> price;
50          }
51      }
```

Row (52 - 69)

```
52         cout << "Enter Page Count of the book: ";
53         cin >> page_count;
54         while (page_count < 0)
55         {
56             cout << "Kindly Enter Positive Page Count: ";
57             cin >> page_count;
58         }
59
60         setdata(page_count, title, price);
61     }
62
63     void putdata()
64     {
65         Publication ::getdata();
66         cout << "Total Pages: " << page_count << endl;
67     }
68 };
69
```

Row (70 - 98)

```
70 class Tape : public Publication
71 {
72 private:
73     float play_time;
74
75 public:
76     void setdata(float p_t, string t, float p)
77     {
78         Publication ::setdata(t, p);
79         play_time = p_t;
80     }
81     void getdata()
82     {
83         string title;
84         float price;
85         cout << "\n--- Enter Tape Details ---" << endl;
86
87         cout << "Enter title of the tape: ";
88         cin.ignore();
89         getline(cin, title);
90
91         cout << "Enter Price of the tape: ";
92         cin >> price;
93         while (price < 0)
94         {
95             cout << "Kindly Enter Positive Price: ";
96             cin >> price;
97         }
98
```

Row (99 - 116)

```
99      cout << "Enter Playtime for the tape: ";
100      cin >> play_time;
101      while (play_time < 0)
102      {
103          cout << "Kindly Enter Positive Playtime: ";
104          cin >> play_time;
105      }
106
107      setdata(play_time, title, price);
108  }
109
110  void putdata()
111  {
112      Publication::getdata();
113      cout << "Total Play Time: " << play_time << " minutes" << endl;
114  }
115  };
116
```

Row (117 - 130)

```
117  int main()
118  {
119      Book b;
120      b.getdata();
121
122      Tape t;
123      t.getdata();
124
125      b.putdata();
126      t.putdata();
127
128      cout << endl;
129      return 0;
130  }
```

Output

```
--- Enter Book Details ---
Enter title of the book: The Obstacle is The Way
Enter Price of the book: 430.750
Enter Page Count of the book: 1350

--- Enter Tape Details ---
Enter title of the tape: Advisory
Enter Price of the tape: 230.50
Enter Playtime for the tape: 15.30

Title: The Obstacle is The Way
Price: 430.75$
Total Pages: 1350

Title: Advisory
Price: 230.5$
Total Play Time: 15.3 minutes
```



Question 02

Code

Row (1 – 25)

```
1  #include <iostream>
2  #include <string>
3  #include <iomanip>
4  using namespace std;
5
6  class Employee
7  {
8  private:
9      string employee_Name;
10     int employee_Number;
11     string hire_Date;
12
13 public:
14     Employee()
15     {
16         employee_Name = "Unknown";
17         employee_Number = 0;
18         hire_Date = "Unknown";
19     }
20     void setdata(string name, int number, string date)
21     {
22         employee_Name = name;
23         employee_Number = number;
24         hire_Date = date;
25     }
```

Row (26 - 53)

```
26     void display()
27     {
28         cout << "\n --- Employee Details --- " << endl;
29         cout << "Employee Name: " << employee_Name << endl;
30         cout << "Employee Number: " << employee_Number << endl;
31         cout << "Hire Date: " << hire_Date << endl;
32     }
33 };
34
35 class ProductionWorker : public Employee
36 {
37 private:
38     int shift;
39     double hourly_Pay;
40
41 public:
42     ProductionWorker() : Employee()
43     {
44         shift = 0;
45         hourly_Pay = 0.0;
46     }
47
48     void setdata(string name, int number, string date, int s, double pay)
49     {
50         Employee::setdata(name, number, date);
51         shift = s;
52         hourly_Pay = pay;
53     }
```

Row (54 - 82)

```
54     void getdata()
55     {
56         string name, date;
57         int number, shift;
58         double pay;
59
60         cout << "\nEnter Employee Name: ";
61         getline(cin, name);
62
63         cout << "Enter Employee Number: ";
64         cin >> number;
65         while (number <= 0)
66         {
67             cout << "Enter a valid Employee Number: ";
68             cin >> number;
69         }
70         cin.ignore();
71
72         cout << "Enter Hire Date (DD-MM-YYYY): ";
73         getline(cin, date);
74
75         cout << "Enter Shift Number (1 or 2): ";
76         cin >> shift;
77         while (shift != 1 && shift != 2)
78         {
79             cout << "Enter a valid Shift Number (1 or 2): ";
80             cin >> shift;
81         }
82     }
```


Row (83 - 109)

```
83         cout << "Enter Hourly Pay Rate: ";
84         cin >> pay;
85         while (pay < 0)
86         {
87             cout << "Enter Positive Hourly Pay Rate: ";
88             cin >> pay;
89         }
90         setdata(name, number, date, shift, pay);
91     }
92     void display()
93     {
94         cout << fixed << setprecision(2) << endl;
95         Employee::display();
96         cout << "Shift: " << shift << endl;
97         cout << "Hourly Pay Rate: $" << hourly_Pay << endl;
98     }
99 };
100
101 int main()
102 {
103     ProductionWorker worker;
104     worker.getdata();
105     worker.display();
106
107     cout << endl;
108     return 0;
109 }
```

Output

```
Enter Employee Name: Alan Ritch
Enter Employee Number: 23415
Enter Hire Date (DD-MM-YYYY): 01-05-2022
Enter Shift Number (1 or 2): 1
Enter Hourly Pay Rate: 45.570
```

```
--- Employee Details ---
Employee Name: Alan Ritch
Employee Number: 23415
Hire Date: 01-05-2022
Shift: 1
Hourly Pay Rate: $45.57
```



Question 03**Code****Row (1 - 25)**

```
1  #include <iostream>
2  #include <string>
3  #include <iomanip>
4  using namespace std;
5
6  class Employee
7  {
8  private:
9      string employee_Name;
10     int employee_Number;
11     string hire_Date;
12
13 public:
14     Employee()
15     {
16         employee_Name = "Unknown";
17         employee_Number = 0;
18         hire_Date = "Unknown";
19     }
20     void setdata(string name, int number, string date)
21     {
22         employee_Name = name;
23         employee_Number = number;
24         hire_Date = date;
25     }
```

Row (26 - 48)

```
26     void display()
27     {
28         cout << "\n --- Employee Details --- " << endl;
29         cout << "Employee Name: " << employee_Name << endl;
30         cout << "Employee Number: " << employee_Number << endl;
31         cout << "Hire Date: " << hire_Date << endl;
32     }
33 };
34
35 class ShiftSupervisor : public Employee
36 {
37 private:
38     double annual_salary;
39     double annual_bonus;
40     double annual_production_goal;
41
42 public:
43     ShiftSupervisor() : Employee()
44     {
45         annual_salary = 0.0;
46         annual_bonus = 0.0;
47         annual_production_goal = 0.0;
48     }
```

Row (49 - 77)

```
49     ShiftSupervisor(string name, int number, string date, double salary, double goal) : Employee()
50     {
51         setdata(name, number, date);
52         annual_salary = salary;
53         annual_bonus = 0.0;
54         annual_production_goal = goal;
55     }
56     void setAnnualSalary(double salary)
57     {
58         annual_salary = salary;
59     }
60     void setAnnualProductionGoal(double goal)
61     {
62         annual_production_goal = goal;
63     }
64     double calculateBonus(double productionAchieved)
65     {
66         // Calculate 10% bonus if production goal is reached
67         if (productionAchieved >= annual_production_goal)
68         {
69             annual_bonus = 0.10 * annual_salary;
70             return annual_bonus;
71         }
72         else
73         {
74             annual_bonus = 0.0;
75             return annual_bonus;
76         }
77     }
```

Row (78 - 103)

```
78     void display()
79     {
80         cout << fixed << setprecision (2) << endl;
81         Employee::display();
82         cout << "Annual Salary: $" << annual_salary << endl;
83         cout << "Annual Bonus: $" << annual_bonus << endl;
84     }
85 };
86
87 int main()
88 {
89     string name, date;
90     int number;
91     double production_Achieved, salary;
92
93     cout << "\nEnter Employee Name: ";
94     getline(cin, name);
95
96     cout << "Enter Employee Number: ";
97     cin >> number;
98     while (number <= 0)
99     {
100         cout << "Enter a valid Employee Number: ";
101         cin >> number;
102     }
103 }
```

Row (104 - 130)

```
104     cin.ignore();
105     cout << "Enter Date of Hire: ";
106     getline(cin, date);
107
108     cout << "Enter Annual Salary: ";
109     cin >> salary;
110     while (salary < 0)
111     {
112         cout << "Enter Valid Annual Salary: ";
113         cin >> salary;
114     }
115
116     cout << "Enter Annual Production (Units) Achieved: ";
117     cin >> production_Achieved;
118     while (production_Achieved < 0)
119     {
120         cout << "Enter Positive Annual Production (Units) Achieved: ";
121         cin >> production_Achieved;
122     }
123
124     ShiftSupervisor supervisor(name, number, date, salary, 10000);
125
126     double bonus = supervisor.calculateBonus(production_Achieved);
127     if (bonus > 0)
128     {
129         cout << "Bonus validated! Annual Bonus: $" << bonus << endl;
130     }
```

Row (131 - 140)

```
131     else
132     {
133         cout << "Production goal not achieved. No bonus granted." << endl;
134     }
135
136     supervisor.display();
137
138     cout << endl;
139     return 0;
140 }
```

Output

```
Enter Employee Name: John Mark
Enter Employee Number: 23415
Enter Date of Hire: 01-02-2021
Enter Annual Salary: 50500.50
Enter Annual Production (Units) Achieved: 10000
Bonus validated! Annual Bonus: $5050.05

--- Employee Details ---
Employee Name: John Mark
Employee Number: 23415
Hire Date: 01-02-2021
Annual Salary: $50500.50
Annual Bonus: $5050.05
```



Question 04

Code

Row (1 - 29)

```
1  ∨ #include <iostream>
2    #include <cstring>
3    #include <iomanip>
4    using namespace std;
5
6  ∨ class Shape
7  {
8    private:
9      string shape_color;
10
11   public:
12   ∨   Shape()
13   {
14       shape_color = "Undefined";
15   }
16   ∨   void setdata(string clr)
17   {
18       shape_color = clr;
19   }
20   ∨   virtual double area()
21   {
22       return -1;
23   }
24   ∨   virtual void display()
25   {
26       cout << "Color: " << shape_color << endl;
27   }
28   };
29
```

Row (30 - 57)

```
30  class Triangle : public Shape
31  {
32  private:
33      double tri_base, tri_height, tri_sides;
34
35  public:
36      Triangle()
37      {
38          tri_base = 0.0;
39          tri_height = 0.0;
40      }
41      void setdata(double b, double h, double ns, string clr)
42      {
43          tri_base = b;
44          tri_height = h;
45          tri_sides = ns;
46          Shape::setdata(clr);
47      }
48      void getdata()
49      {
50          double base, height, sides;
51          string color;
52
53          cout << "\nEnter Color of Triangle: ";
54          getline(cin, color);
55
56          cout << "Enter base of triangle: ";
57          cin >> base;
```


Row (58 - 81)

```
58         while (base < 1)
59         {
60             cout << "Kindly Enter Positive Base: ";
61             cin >> base;
62         }
63
64         cout << "Enter Height of Triangle: ";
65         cin >> height;
66         while (height < 0)
67         {
68             cout << "Kindly Enter Positive Height: ";
69             cin >> height;
70         }
71
72         cout << "Enter Number of Sides of Triangle: ";
73         cin >> sides;
74         while (sides < 0)
75         {
76             cout << "Kindly Enter Positive Sides: ";
77             cin >> sides;
78         }
79
80         setdata(base, height, sides, color);
81     }
```

Row (82 - 110)

```
82     double area() override
83     {
84         return (0.5 * tri_base * tri_height);
85     }
86     void display() override
87     {
88         cout << fixed << setprecision(2) << endl;
89         cout << "\n --- Triangle Details --- " << endl;
90         cout << "Base: " << tri_base << endl;
91         cout << "Height: " << tri_height << endl;
92         cout << "Number of Sides: " << tri_sides << endl;
93         Shape::display();
94         cout << "Area: " << area() << endl;
95     }
96 };
97
98 class Circle : public Shape
99 {
100 private:
101     double radius, center_X, center_Y;
102
103 public:
104     void setdata(double r, double cX, double cY, string clr)
105     {
106         radius = r;
107         center_X = cX;
108         center_Y = cY;
109         Shape::setdata(clr);
110     }
```

Row (111 - 139)

```
111 void getdata()
112 {
113     double rad, centerX, centerY;
114     string color;
115
116     cout << "\nEnter Color of Circle: ";
117     cin.ignore();
118     getline(cin, color);
119
120     cout << "Enter Radius of Circle: ";
121     cin >> rad;
122     while (rad < 0)
123     {
124         cout << "Kindly Enter Positive Radius: ";
125         cin >> rad;
126     }
127
128     cout << "Enter Center X Coordinate of Circle: ";
129     cin >> centerX;
130
131     cout << "Enter Center Y Coordinate of Circle: ";
132     cin >> centerY;
133
134     setdata(rad, centerX, centerY, color);
135 }
136 double area() override
137 {
138     return (3.14 * radius * radius);
139 }
```

Row (140 – 161)

```
140     double circumference()
141     {
142         return (2 * 3.14 * radius);
143     }
144     double diameter()
145     {
146         return (2 * radius);
147     }
148     void display() override
149     {
150         cout << fixed << setprecision(2) << endl;
151         cout << "\n --- Circle Details --- " << endl;
152         cout << "X Coordinate: " << center_X << endl;
153         cout << "Y Coordinate: " << center_Y << endl;
154         cout << "Radius: " << radius << endl;
155         Shape::display();
156         cout << "Area: " << area() << endl;
157         cout << "Diameter: " << diameter() << endl;
158         cout << "Circumference: " << circumference() << endl;
159     }
160 };
161
```

Row (161 - 175)

```
162     int main()
163     {
164         Triangle t;
165         t.getdata();
166
167         Circle c;
168         c.getdata();
169
170         t.display();
171         c.display();
172
173         cout << endl;
174         return 0;
175     }
```

Output

```
Enter Color of Triangle: Red
Enter base of triangle: 5.5
Enter Height of Triangle: 7.5
Enter Number of Sides of Triangle: 3

Enter Color of Circle: Blue
Enter Radius of Circle: 3.5
Enter Center X Coordinate of Circle: 1
Enter Center Y Coordinate of Circle: 3

--- Triangle Details ---
Base: 5.50
Height: 7.50
Number of Sides: 3.00
Color: Red
Area: 20.62

--- Circle Details ---
X Coordinate: 1.00
Y Coordinate: 3.00
Radius: 3.50
Color: Blue
Area: 38.47
Diameter: 7.00
Circumference: 21.98
```

THE END
